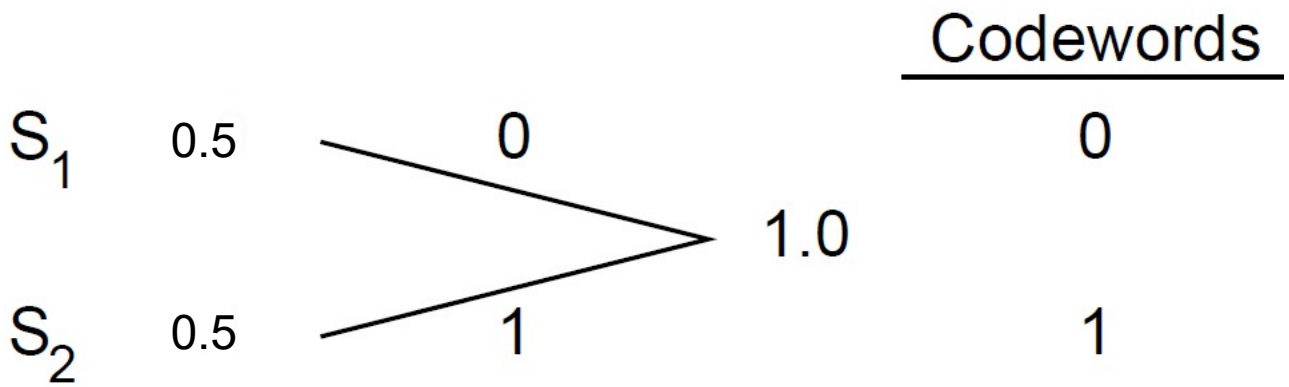
Block Huffman Coding

Realize the limitation of Huffman coding



$$H = 0.5 \log_2\left(\frac{1}{0.5}\right) + 0.5 \log_2\left(\frac{1}{0.5}\right) \approx 1 \text{ bit}$$

Average code length = 0.9 x 1 + 0.1 x 1 = 1

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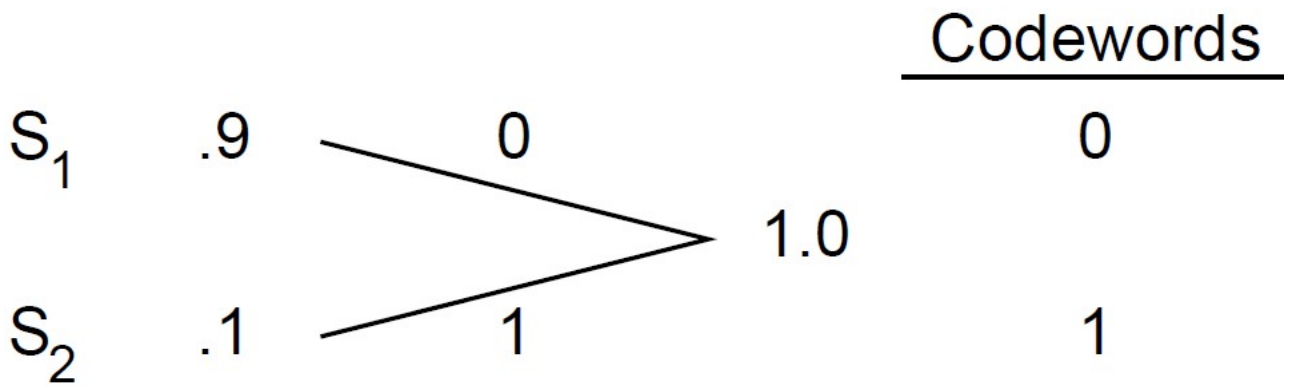


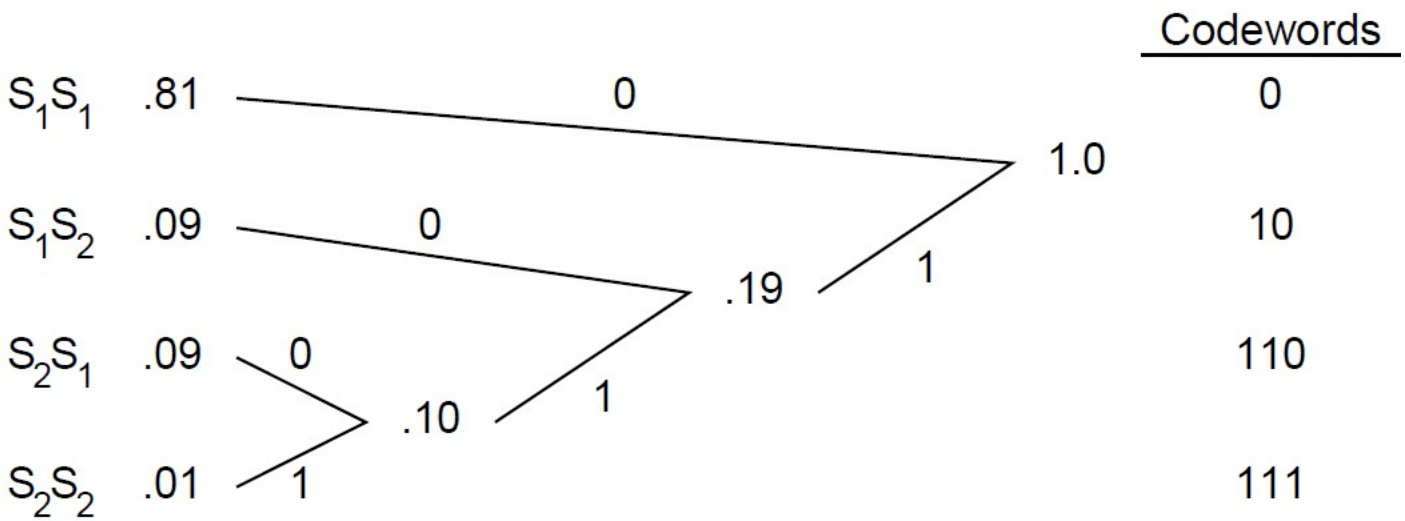
$$H = 0.9 \log_2 \left(\frac{1}{0.9} \right) + 0.1 \log_2 \left(\frac{1}{0.1} \right) \approx 0.47 \ \text{ bits/symbol}$$

Average code length = 0.9 x 1 + 0.1 x 1 = 1

Make blocks

- Take 2 bits at a time
- Possible combinations: s_1s_1, s_1s_2, s_2s_1, and s_2s_2
- Corresponding PDF:
- {0.9 x 0.9, 0.9 x 0.1, 0.1 x 0.09, 0.1 x 0.1}
- {0.81, 0.09, 0.09, 0.01}

Construct Huffman coding for the blocks



<u>Average code per block</u>

$$1(0.81) + 2(0.09) + 3(0.09 + 0.01) = 1.29.$$

<u>Average code per symbol</u>

1.29/2 = 0.645 bits/symbol

Observations

- 2 symbols with PDF {0.9, 0.1}
- Using blocks of length 2 helps
- 4 patterns (i.e. blocks) with PDF {0.81, 0.09, 0.09, 0.01}
- Reduces average codeword length from 1 bit/symbol to 0.645 bits/symbol
- Entropy is 0.47
- Coding efficiency = 0.47/0.645 = 72.8%

Generalize blocks

- M symbols
- k strings in the blocks
- How many blocks? Answer : M^k
- Generate PDF for blocks
- Do the Huffman for the blocks

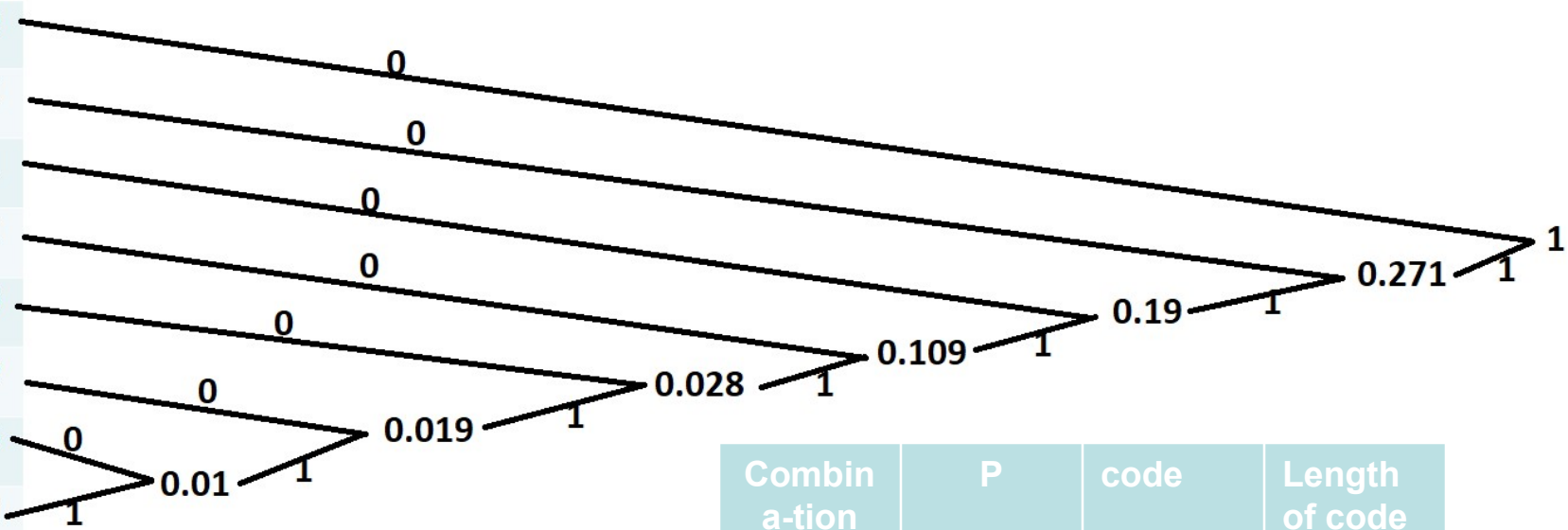
Make blocks

- Take 3 bits at a time
- Possible combinations: 8
- Entropy=1.407 bits

Combina -tion	P	
s1s1s1	0.729	
s1s1s2	0.081	
s1s2s1	0.081	
s1s2s2	0.009	
s2s1s1	0.081	
s2s1s2	0.009	
s2s2s1	0.009	
s2s2s2	0.001	

Combina -tion	P	
s1s1s1	0.729	
s1s1s2	0.081	
s1s2s1	0.081	
s2s1s1	0.081	
s1s2s2	0.009	
s2s1s2	0.009	
s2s2s1	0.009	
s2s2s2	0.001	

Combina -tion	P
s1s1s1	0.729
s1s1s2	0.081
s1s2s1	0.081
s2s1s1	0.081
s1s2s2	0.009
s2s1s2	0.009
s2s2s1	0.009
s2s2s2	0.001



$$Avergare\ Length = (1\ X\ 0.729) + (2\ X\ 0.081) + (3\ X\ 0.081) + (4\ X\ 0.081 + (5\ X\ 0.009) + (6\ X\ 0.009) + (7\ X\ 0.009) + (7\ X\ 0.001)$$

= 1.627

Coding efficiency = 1.407/1.627 = 86.5%

Combin a-tion	P	code	Length of code
s1s1s1	0.729	0	1
s1s1s2	0.081	10	2
s1s2s1	0.081	110	3
s2s1s1	0.081	1110	4
s1s2s2	0.009	11110	5
s2s1s2	0.009	111110	6
s2s2s1	0.009	1111110	7
s2s2s2	0.001	1111111	7